Daily Drilling Report
SPD19B-03
9/1/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
36.0	12 1/4"	181.00	72.83	31.70	99.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
13 3/8" Shoe, 1,487.00mKB	220,877.90	7,730.44	8,416,696.84		31.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,694.00	2,725.00		31.00	8.53	3.6

Operations at Report Time
POOH BHA#9 @ 290m.

Operations Summary
Drilling 12 1/4'' hole section f/ 2694m to 2725m (Section TD). Sweep hole. Wiper trip to 2437m. POOH to 2350m. Run back to bottom. Sweep hole. Spot 15 ppg weighted pill. POOH f/ 2725m to 2200m.

Operations next 24hrs
Cont' POOH BHA#9 to surface. Retrieve wear bushing. Wash inside WH. Run 9 5/8'' casing.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Hamed Daneshkhah / Moslem Yarahmadi	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	8/31/2014	10/1/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	9/1/2014	11/1/2014
Pit Drill	8/25/2014	10/10/2014
Trip Drill	8/28/2014	10/18/2014

STOP Cards

Number of Reports: 36

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	10:30	10....	12 1/4"	D	03	P	Directional drilling 12 1/4" hole section in rotation mode f/ 2694m to 2725m (Section TD) w/ 800 gpm/ 2200-2300 psi/ 40 kips/ 60 rpm/ 10-25 TRQ. ** Drilling with NL/NG. ** Pump 20 bbl Hi-Vis pill when STD down. ** B/R and ream down each stand drilled once. ** Take survey before connection. >> Drilling procedure in Aghar shale member (2675.7m - 2703m): * For the first 10m in Aghar, B/R gently to top of Aghar every 1m drilled. * For the rest of drilling in Aghar, B/R gently to top of Aghar every 5m drilled. * At stand down, pump 30 bbl Lo-Vis + 30 bbl Hi-Vis pill, B/R and R/D entire stand twice. * Spot 20 bbl 15 ppg weighted pill around BHA prior to connection. * Skip survey in Aghar.
10:30	12:00	1.50	12 1/4"	D	07	P	Sweep hole by pumping 30 bbl Lo-Vis + 30 bbl Hi-Vis pill. Circulate w/ 850 gpm/ 2300 psi/ 30 rpm till shakers clean.
12:00	17:00	5.00	12 1/4"	D	06	P	Perform wiper trip by normal B/R f/ 2725m to 2437m w/ 700 gpm/ 1650 psi/ 40 rpm/ 5-16 TRQ. Hole in good condition.
17:00	17:30	0.50	12 1/4"	D	04	P	POOH on elevator f/ 2437m to 2350m and check hole without parameters, OK.
17:30	19:30	2.00	12 1/4"	D	04	P	Run back f/ 2350m to bottom (2725m) on elevator in good hole condition.
19:30	21:30	2.00	12 1/4"	D	07	P	Sweep hole by pumping 20 bbl Lo-Vis + 20 bbl Hi-Vis pill, circulate 30 min, pump 20 bbl Lo-Vis + 20 bbl Hi-Vis pill, circulate w/ 850 gpm/ 2300 psi/ 30 rpm, till shakers clean.
21:30	22:00	0.50	12 1/4"	D	09	P	Spot 130 bbls 15 ppg weighted pill (Cover f/ Btm to top of S-8 Member).
22:00	00:00	2.00	12 1/4"	D	04	P	POOH directional BHA#9 f/ 2725m to 2200m. Hole in good condition.
00:00	00:00						Sep 02 , 2014 00:00 - 06:00 Cont' POOH directional BHA#9 f/ 2200m to 290m.

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PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,725.00	KCl-Polymer Mud	12.10	74	30.0	32.000	2.9
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
9.5	0.200	1.500	0.800			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
12	14	2.0	75.0	23.0	58,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
11.000	22.000	28.000	2.6/97.4		1937.0	1250.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
5266.0	109.0	0.0			38.28134	0.289

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.10	6.	300	2.0	996.0	34.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Sea Command	Supply Boat	9/1/2014 15:30		Del all deck Cargo, Emb 02 PAX and proceed to Sahar-2 to bring Cargo for DD6.
Swissco	Supply Boat	8/31/2014 02:30	9/1/2014 18:50	Received Kish backloads. C/Off from Rig and departed for Aban-8.
Marina-1	Supply Boat	8/16/2014 15:15		Stand by Boat.

Mud Additives

Description	Consumed	Cum Rec	On Loc
CaCo3-F BB		159.0	72.0
Calcium Carbonate Coarse	1.0	3.0	2.0
Barite Bulk	25.0	249.0	137.0
Starch		1,200.0	217.0
CaCo3-M BB	1.0	4.0	3.0
Caustic Soda 2014	10.0	340.0	170.0
Spersene	5.0	145.0	120.0
Pac-LV	21.0	481.0	135.0
Soda Ash 25 Kg	30.0	550.0	197.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	0.0
Cement G	0.0	0.0	179.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0
Diesel Fuel	14.0	0.0	185.0
Barite Bulk	25.0	0.0	137.0
Drill Water	25.0	0.0	275.0
Potable Water	23.0	20.0	99.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
9	12 1/4	Varel	VM616PGUX-V2	6011971	PDC	M422	12/12/12/12/12/12/12/12
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.99	55.68	486,386.40	2,437.00	2,725.00	288.00	5.2	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
750	810	1,650.0	2,300.0	7	40	125	157
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
81.56	55.68	100.00	0.00	410	364	296	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
2,694.00	2,725.00	8.53	22,000.0	800	60	2,300.0	40	364

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.000	27.83	231.39
7" Drilling Jar (COR-3022-AD)	7	2.938	5.85	203.56
16 jts 5 7/8" HWDP	5 7/8	4.000	149.23	197.71
X/O	7	2.813	0.79	48.48
1 jt 8 1/4" DC	8 1/4	2.810	9.46	47.69
8 1/2" Float Sub	8 1/2		0.79	38.23
8" NMDC	8	3.500	9.44	37.44
8" TM - Sub	8		2.03	28.00
8" NMDC	8	3.500	9.46	25.97
11 7/8" S. Stabilizer (COR3103AD)	11 7/8	2.813	2.16	16.51
8" PNMDC	8 1/16	3.500	4.03	14.35
9 5/8" Motor (AKO: 1.00° , Sleeve: 12 1/8")	9 5/8		9.95	10.32
12 1/4" PDC bit	12 1/4		0.37	0.37

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4

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Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
2,701.40	249.00	21.10	2,558.05	-338.14	-739.61	813.24	0.19	0.11

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
S4		2,594.00	2,457.78
S3		2,622.00	2,483.90
S2		2,632.50	2,493.72
S1		2,675.70	2,534.06
Upper Kangan		2,703.00	

General Remarks

Comment

>> Requirements: (MI MR#14) + (HPOGC 6 3/4" Motor)

>>> Sent:

** Swissco to Kish: (COR String Stabilizers: 1 NO 12" COR3059AD + 1 NO 11 7/8" COR2458AD + 1 NO 11 3/4" COR3053AD + 1 NO 11 7/8" COR3056AD + 1 NO 12 1/8" COR2863AD + 1 NO 12 1/8" OMM-3111 + 1 NO 12" COR3060AD + 1 NO 11 3/4" COR2885AD + 1 NO 11 1/2" COR3052AD + 1 NO 12" COR2638AD) + (PPL 12 1/4" PDC BITS: 2 EA VRP716P + 3 EA VM813GXU + 5 EA VM616PDGUX_V1 + 1 EA VMPGUX_V1 + 1 EA VM616PDGUX_V2 + 3 EA VM813GXU + 1 EA VRP716P + 3 EA VM616PGUX_V1 + 3 EA VMPGUX_V2 + 3 EA VRP813GU + 2 EA VM819GUX + 2 EA BEST BITS, M1385) + (1 NO 12 1/4" INSERT BIT, CH04JMRSV + 1 NO 12 1/4" INSERT BIT, CH34DMRSV) + (COR Fishing tools: 1# Basket (SN: EOE-8MSB-01) Containing: (17 1/2", 16", 12 1/4") Junk Mills,14" Junk Sub, 9 5/8" JUNK Sub, 11" & 15" Reverse Circulating Junk Basket, 6 3/8" Taper Tap, 8 1/4" Casing Spear, 2# Spear Stop Sub w/ 15" & 20.5" Stop Rings, 15" & 20" Impression Block, 8" Safety Joint, Marine Spear for 13 3/8" to 20"Casing w/Spear Grapple & Integral Stop Ring, Spear Grapple for 18 5/8" Casing, Spear Grapple for 13 3/8" casing,7 3/4" Bumper Sub, 10 1/2" & 15" (with Guide) Magnet.) + (COR Fishing tools: 1 no 7 3/4" Fishing Accelerator Jar) + (Garbage).

>>> Received:

** Sea Command f/ Kish: (MI Chemicals: 70 BB CaCO3 - F) + (AMOS Cementing tools: 2 no 10 3/4" Double cement head with related XOS + 1 no 10 3/4" Swage, BTC pin + 1 no 10 3/4" Swage, Vam Top pin + 1 no 2" Hammer valve + 1 no 4" Butterfly valve + 4 1/2" Packing) + (AL-Ghaith CSG handling tools: 2 no Power Unit + 2 no Power Tong with 9 5/8" and 10 3/4" Jaws + 2 no 9 5/8" Spider Elevator + 1 no 10 3/4" Spider Elevator + 1 no 9 5/8" FMS + 1 no 10 3/4" FMS + 2 no 10 3/4" Hand Slips + 2 no 9 5/8" hand Slips + 2 no 9 5/8" Side Door Elevator + 2 no 10 3/4" Side Door Elevator + 2 no 9 5/8" Single Joint Elevator + 2 no 10 3/4" Single Joint Elevator + 1 no 9 5/8" Safety Clamp + 1 no 10 3/4" Safety Clamp + ...

Comment

..... + 4 no 10 3/4" Thread Protector + 4 no 9 5/8" Thread Protector + 1 no 10 3/4" Fill up tools + 1 no Combination 9 5/8" - 10 3/4" Fill up tools + 2 no Jam Unit without Dump Valve + 1 no 9 5/8" Stabbing guide + 1 no 10 3/4" Stabbing guide + 1 no 9 5/8" Drift + 1 no 10 3/4" Drift + 1 no 9 5/8" Manual tong + 1 no 10 3/4" Manual tong) + (HPOGC Tools: 4 no 6 1/2" NMDC + 2 no 6 1/2" PNMDC + 2 no 6 1/2" Float Sub + 2 no 6 1/2" TM - Sub + 2 no 6 1/2" Restrictor Sub + 2 no 4 1/2" IF APC Protector + 2 no MWD Transmitter + 5 no 5R Float Valve + 1 no 4 1/2" IT Lifting Sub) + (8 1/2" Bits: 4 no PDC, VM713GUX + 1 no PDC, VM616GUX + 1 no Insert, CH18MRS + 1 no Mill Tooth, CH4GMS) + (1 no 6" Mill Tooth Bit) + (1 no 6" Insert Bit) + (2 no 5 7/8" PDC Bit, VRP613PDGX) + (2 no Spacer for TBG HGR load shoulder) + (COR IBS Stabilizers: 1 no 8 3/8" + 2 no 8 1/8" + 2 no 8" + 2 no 7 7/8" + 1 no 7 3/4" + 1 no 7 1/2" + 1 no 7 1/4") + (COR Drilling Jars: 1 no 7" DA Cougar Jar + 1 no 6 1/2" HM NOV Jar) + (COR Fishing Tools, 25 Items : 5 7/8" Junk Mill + 6 1/2" Bumper Sub + 3 3/4" Taper Tap + 5 3/4" Casing Spear + 6.220" CSG Spear Grapple + 6.095" CSG Spear Grapple + 8" Impression Block + 5 1/2" Impression Block + 6 1/2" Safety Joint + 8 1/8" FS Overshot + 36" Long Extension for 8 1/8" Overshot + 8 1/8" SH Overshot) + (Basket Grapples for 8 1/8" FS Overshot: 5 1/2" , 5 1/4" , 5" , 4 3/4" , 3 1/2" , 3 3/8" , 5 1/2" , 4 3/4") + (Spiral Grapples for 8 1/8" FS Overshot: 7" , 6 7/8" , 6 5/8" , 6 3/4" , 6 1/2").